Question 2:

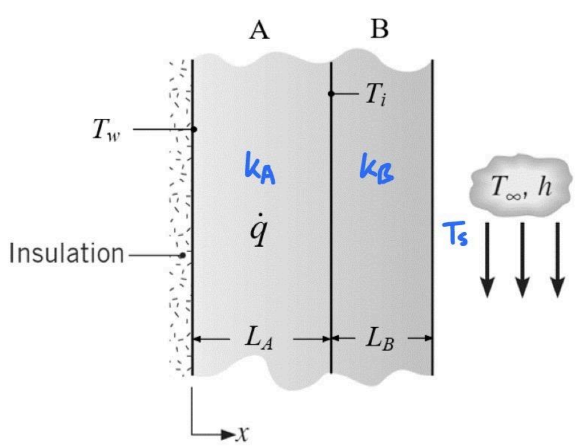
Known:

- **Uniform volumetric heat generation rate = qdot**
- **The left surface of Wall A is thermally insulated**
- **There is no heat generation within Wall B**
- **The left surface of Wall B is in contact with Wall A**
- **There is no contact thermal resistance at their boundary**
- **Coolant temperature: Tinfinity = 20 C**
- **Temperature between insulation and Wall A: Tw = 115 C**
- **Temperature between Wall A and Wall B: Ti = 100 C**

Find:

- **(A) Calculate the right surface temperature Ts of Wall B**
- **(B) Calculate the uniform volumetric heat generation qdot at steady state**
- **(C) Calculate the thermal conductivity kA of Wall A**
- **(D) Plot the temperature profile across Wall A and Wall B and show its important features**

Schematic:



Properties:

- **LA = 30 cm**
- **Thermal conductivity of Wall A = kA**
- **LB = 10 cm**
- **Thermal conductivity of Wall B: kB = 30 W/(m*K)**
- **Convective heat transfer coefficient: h = 500 W/(m^2*K)**

Assumptions:

- **Assume 1-D heat transfer**
- **Assume constant properties**

Analysis:

ⓐ $\dot{E}_{in} + \cancel{\dot{E}_G}^{\,0} - \dot{E}_{out} = \cancel{\dot{E}_s}^{\,0} \longrightarrow \dot{E}_{in} = \dot{E}_{out}$

$\dot{E}_{in} = k_B \frac{(T_s - T_i)}{L_B}$; $\dot{E}_{out} = h(T_s - T_\infty)$

$k_B \frac{(T_s - T_i)}{L_B} = h(T_s - T_\infty) \longrightarrow (30\ W/mK)\frac{(T_s - 100°C)}{0.1m} = (500\ W/m^2K)(T_s - 20°C) \longrightarrow T_s = 50°C$

ⓑ $\cancel{\dot{E}_{in}}^{\,0} + \dot{E}_G - \dot{E}_{out} = \cancel{\dot{E}_s}^{\,0} \longrightarrow \dot{E}_G = \dot{E}_{out}$

$\dot{E}_G = \dot{q}V$; $\dot{E}_{out} = h(T_s - T_\infty)$

$\dot{q}V = h(T_s - T_\infty) \longrightarrow \dot{q} = \frac{(500\ W/m^2K)(50°C - 20°C)}{(0.3m)} = 50{,}000\ W/m^3$

ⓒ $\int \frac{d^2T}{dx^2} = \int \frac{-\dot{q}}{k_A} \longrightarrow \frac{dT}{dx} = \frac{-\dot{q}}{k_A}x + C_1$ $\left.\frac{dT}{dx}\right|_{x=0} = 0 \longrightarrow C_1 = 0$

$\int \frac{dT}{dx} = \int -\frac{\dot{q}}{k_A}x + C_1 \longrightarrow T(x) = \frac{-\dot{q}}{2k_A}x^2 + C_1x + C_2$

$T(0) = T_o = 115 = C_2$

$T(L_A) = T_i = -\frac{\dot{q}}{2k_A}L_A^2 + C_2 \longrightarrow 100 = -\frac{50000}{2k_A}(0.3)^2 + 115 \longrightarrow k_A = 150\ W/mK$

ⓓ $T_A(x) = 115 - \frac{\dot{q}}{2k_A}x^2$

Wall B:

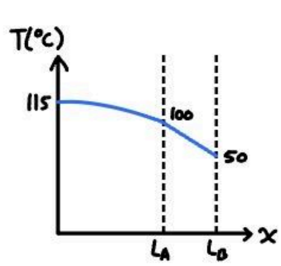
$\frac{d^2T}{dx^2} = 0$

$T_B(x) = C_1x + C_2$

$T_B(L_A) = 100 = C_1L_A + C_2$

$T_B(L_A + L_B) = 50 = C_1(L_A + L_B) + C_2 \longrightarrow C_1 = -500$; $C_2 = 250$

$T_B(x) = 250 - 500x$



Question 3:

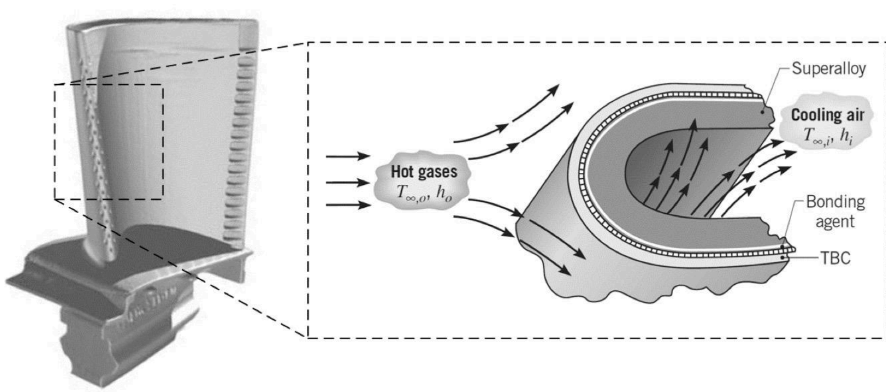
Known:

- **External hot gases: Tinfinity,o = 1616 K**
- **Cooling air: Tinfinity,i = 400 K**
- **The turbine blade is at steady state**
- **The turbine blade can be approximated as a plane wall**
- **The maximum allowable temperature for the superalloy is Tmax = 1200 K**

Find:

- **(A) If we do not apply the TBC and directly expose the superalloy blade to hot gases, draw the thermal resistance network for this condition and label all thermal resistances. Calculate the exterior surface temperature of the superalloy blade. Can the superalloy blade be maintained below Tmax?**
- **(B) Now we apply LTBC = 0.5 mm thick TBC to the superalloy blade. Draw the thermal resistance network for this condition and label all thermal resistances. Calculate the temperature of the superalloy blade again. Can the superalloy blade be maintained below Tmax this time?**

Schematic:



Properties:

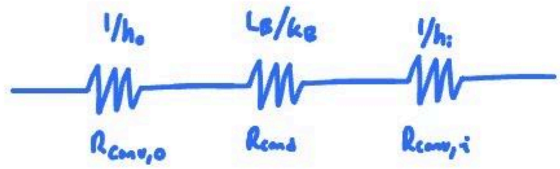
- **Superalloy thermal conductivity: kb = 25 W/(m*K)**
- **Ceramic thermal conductivity: kTBC = 1 W/(m*K)**
- **Interfacial thermal resistance: R" = 10^-4 (m^2*K)/W**
- **Heat transfer coefficient of hot gases: ho = 1000 W/(m^2*K)**
- **Heat transfer coefficient of cooling air: hi = 500 W/(m^2*K)**
- **Thickness of superalloy blade: Lb = 5 mm**

Assumptions:

- **Assume thermal radiation can be neglected**
- **Assume one-dimensional heat conduction through the TBC and Superalloy**

Analysis:

a



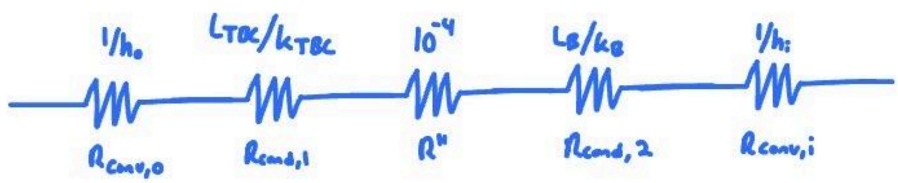
$$R_{tot} = \frac{1}{h_o} + \frac{L_B}{k_B} + \frac{1}{h_i} = \frac{1}{100} + \frac{0.005}{25} + \frac{1}{500} = 0.0032 \, m^2K/W$$

$$q = \frac{T_{\infty,o} - T_{\infty,i}}{R_{tot}} = \frac{1616 - 400}{0.0032} = 3.8 \times 10^5 \, W/m^2K$$

$$q = h_o(T_{\infty,o} - T_{s,o}) \longrightarrow 3.8 \times 10^5 = 1000(1616 - T_{s,o})$$

$T_{s,o} = 1236K$ above T_{max}

b



$$R_{tot} = \frac{1}{100} + \frac{0.0005}{1} + 10^{-4} + \frac{0.005}{25} + \frac{1}{500} = 0.0038 \, m^2K/W$$

$$q = \frac{1616 - 400}{0.0038} = 3.2 \times 10^5 \, W/m^2K$$

$$q = h_o(T_{\infty,o} - T_{s,o})$$

$$T_{s,o} = T_{\infty,o} - \frac{q}{h_o} = 1296k$$

$$T_{s,1} = T_{s,o} - q \cdot L_{TBC} = 1136k$$

$T_{s,2} = T_{s,1} - q \cdot R'' = 1104k$ below T_{max}